

Homophily in Congress on Twitter

Marcel Julien Kalb^{*†} Mike Barth^{*‡} Sam Känner^{*§}

Introduction

We examine the Twitter Interaction Network for the 117th U.S. Congress by SNAP. The dataset contains 475 congressional Twitter accounts connected by 13,289 directed, weighted edges representing retweets, quote tweets, replies, and mentions. Edge weights capture the normalized influence between accounts as transmission probabilities. As the dataset contains no node attributes beyond usernames, we add party affiliation, chamber, age, and ideological scores from [Voteview](#), [github/unitedstates](#), and [USCD](#). We assess homophily across these characteristics, with particular interest in whether chamber and age moderate partisan homophily.

We assess homophily across these characteristics. It has been shown thoroughly that party homophily exists, but we are most interested in whether tertiary factors such as chamber and age moderate this effect, e.g. do members of the senate exhibit more homophily than those in the house, and, do older members exhibit more or less homophily than their younger colleagues?

Theory

Homophily is the principle that contact between similar people occurs at a higher rate than among dissimilar people. It can be further distinguished in two types: *status homophily* as the similarity of persons in their formal, informal or ascribed status and *value homophily* based on shared beliefs, values and attitudes of persons (McPherson, Smith-Lovin, and Cook 2001). We take the party affiliation, the chamber and the age as status homophily and expect that the number of ties between members of congress is shaped by their status, e.g. congress members mostly interact with members of their own party. Age is expected to further shape the number of connections, since people tend to form ties with those of similar life stage (McPherson, Smith-Lovin, and Cook

^{*}Department of Politics and Public Administration, University of Konstanz.

[†]Email: marcel.julien.kalb@uni-konstanz.de.

[‡]Email: mike.barth@uni-konstanz.de.

[§]Email: sam.kaenner@uni-konstanz.de.

2001), older legislators may have formed cross-party relationships before the current era of polarization, but also older members may tend to be less active on twitter. Also a legislators chamber structures the opportunity for interaction, as Senate and House members operate in largely separate institutional environments with different rules, schedules, and legislative responsibilities, making cross-chamber ties less likely to form regardless of party. Value homophily is captured as left-right ideological position and stance on social and racial issues. There is a considerable tendency for adults to associate with those of their own political orientations (McPherson, Smith-Lovin, and Cook 2001). Therefore, we expect to find more interactions between members with similar positioning in the ideological dimensions.

Assortativity is the network-level measure that formally captures homophily. Newman (2003) defines the assortativity coefficient as a Pearson correlation computed over all edges, ranging from -1 to 1 , where 1 indicates that all edges are within-group. In the context of the congressional Twitter network, assortativity provides a useful way to evaluate whether interaction patterns reflect political and demographic divisions. We therefore use assortativity to assess whether members cluster by party affiliation, chamber, age, and ideological position, and to compare the relative strength of these different forms of homophily within the network.

Data Pre-Processing

i Using `"',"` as decimal and `"'."` as grouping mark. Use ``read_delim()`` for more control.

```
# investigate NA values
vdf <- igraph::as_data_frame(g, what = "vertices")
na_counts <- colSums(is.na(vdf))

data.frame(
  `NA count` = na_counts,
  `NA %`     = round(na_counts / nrow(vdf) * 100, 1),
  check.names = FALSE
) |>
  arrange(desc(`NA count`))
```

	NA count	NA %
district	92	19.4
chamber	47	9.9
party	47	9.9
state	47	9.9
bioguide_id	47	9.9
icpsr	47	9.9
nominate_dim1	47	9.9
nominate_dim2	47	9.9

born	47	9.9
age24	47	9.9
full_name	5	1.1
name	0	0.0
username	0	0.0

Exploratory Data Analysis

Overall, the network is moderately sized and active, but not densely connected. Members form many retweet ties in absolute terms, yet these ties represent only a small share of all possible member-to-member connections ($\approx 5\%$). The typical congress member has around 22–24 connections, while a small number of members have far higher connectivity. This suggests a selective and uneven retweet structure, with activity concentrated around more central or more active members rather than spread uniformly across the network. This is to be expected given in politics there are often central figures that may receive more attention, such as presidents or speakers.

```
# Core descriptives: graph size

n_vertices <- vcount(g)
n_edges    <- ecount(g)
density    <- edge_density(g)

size_summary <- tibble::tibble(
  Measure = c(
    "Number of vertices",
    "Number of edges",
    "Edge density"
  ),
  Value = c(
    n_vertices,
    n_edges,
    density
  )
) %>%
  dplyr::mutate(Value = round(Value, 3))

cat(
  knitr::kable(
    size_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
```

```
)
```

Measure	Value
Number of vertices	475.000
Number of edges	13289.000
Edge density	0.059

```
# Degree and weighted degree / strength
```

```
deg_in <- degree(g, mode = "in")
deg_out <- degree(g, mode = "out")
```

```
str_in <- strength(g, mode = "in")
str_out <- strength(g, mode = "out")
```

```
degree_summary <- tibble::tibble(
  Measure = c(
    "In-degree",
    "Out-degree",
    "Weighted in-strength",
    "Weighted out-strength"
  ),
  Min = c(
    min(deg_in, na.rm = TRUE),
    min(deg_out, na.rm = TRUE),
    min(str_in, na.rm = TRUE),
    min(str_out, na.rm = TRUE)
  ),
  Q1 = c(
    quantile(deg_in, 0.25, na.rm = TRUE),
    quantile(deg_out, 0.25, na.rm = TRUE),
    quantile(str_in, 0.25, na.rm = TRUE),
    quantile(str_out, 0.25, na.rm = TRUE)
  ),
  Median = c(
    median(deg_in, na.rm = TRUE),
    median(deg_out, na.rm = TRUE),
    median(str_in, na.rm = TRUE),
    median(str_out, na.rm = TRUE)
  ),
  Mean = c(
    mean(deg_in, na.rm = TRUE),
    mean(deg_out, na.rm = TRUE),
    mean(str_in, na.rm = TRUE),
    mean(str_out, na.rm = TRUE)
  )
)
```

```

    mean(str_out, na.rm = TRUE)
  ),
  Q3 = c(
    quantile(deg_in, 0.75, na.rm = TRUE),
    quantile(deg_out, 0.75, na.rm = TRUE),
    quantile(str_in, 0.75, na.rm = TRUE),
    quantile(str_out, 0.75, na.rm = TRUE)
  ),
  Max = c(
    max(deg_in, na.rm = TRUE),
    max(deg_out, na.rm = TRUE),
    max(str_in, na.rm = TRUE),
    max(str_out, na.rm = TRUE)
  )
) %>%
dplyr::mutate(
  dplyr::across(where(is.numeric), ~ round(.x, 3))
)

cat(
  knitr::kable(
    degree_summary,
    format = "pipe"
  ),
  sep = "\n"
)

```

Measure	Min	Q1	Median	Mean	Q3	Max
In-degree	0.000	13.000	22.000	27.977	37.000	127.000
Out-degree	1.000	17.000	24.000	27.977	35.000	210.000
Weighted in-strength	0.000	0.058	0.117	0.163	0.204	1.648
Weighted out-strength	0.002	0.080	0.134	0.163	0.196	0.944

```

# Degree distributions

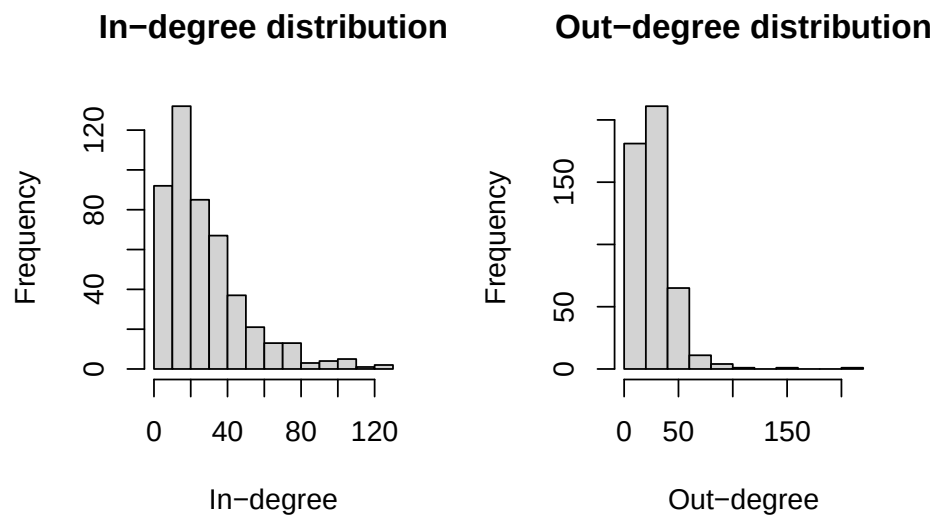
par(mfrow = c(1, 2))

hist(
  deg_in,
  main = "In-degree distribution",
  xlab = "In-degree"
)

hist(

```

```
deg_out,
main = "Out-degree distribution",
xlab = "Out-degree"
)
```



```
par(mfrow = c(1, 1))

# Structural network measures

recip <- reciprocity(g)

trans_global <- transitivity(g, type = "global")
trans_avg <- transitivity(g, type = "average")

assort_degree <- assortativity_degree(g, directed = TRUE)

structure_summary <- tibble::tibble(
  Measure = c(
    "Reciprocity",
    "Global transitivity",
    "Average local transitivity",
    "Degree assortativity"
  ),
  Value = c(
    recip,
    trans_global,
    trans_avg,
    assort_degree
  )
)
```

```

)
) %>%
  dplyr::mutate(Value = round(Value, 3))

cat(
  knitr::kable(
    structure_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
)

```

Measure	Value
Reciprocity	0.462
Global transitivity	0.270
Average local transitivity	0.301
Degree assortativity	-0.096

The network shows a moderate level of reciprocity: about 46% of directed ties are reciprocated. This suggests that interactions are not purely one-sided, but often occur between pairs of members who engage with each other, possibly congress members working on the same topics or from similar geographic regions. Transitivity is also moderate, with global and average local clustering around 0.27–0.30, indicating that interaction patterns are somewhat clustered within groups of mutually connected members. This is to be expected given we are not accounting for chamber affiliation and or party affiliation at the moment, so clustering is to be expected. The slightly negative degree assortativity (-0.096) suggests that members with many outgoing interactions are not especially likely to direct those interactions toward members with many incoming interactions. In other words, highly active congress members are not disproportionately interacting with highly visible or frequently targeted congress members.

```

# Homophily by party
e1 <- igraph::as_data_frame(g, what = "edges")
e1$from_party <- V(g)$party[match(e1$from, V(g)$name)]
e1$to_party <- V(g)$party[match(e1$to, V(g)$name)]
mix <- xtabs(weight ~ from_party + to_party, data = e1)
mix_prop <- mix / rowSums(mix)
cat("Overall\n", knitr::kable(round(mix_prop, 3), format =
  ↵ "pipe"), sep = "\n")

```

Overall

	Democrat	Independent	Republican

	Democrat	Independent	Republican
Democrat	0.874	0.001	0.125
Independent	0.985	0.000	0.015
Republican	0.123	0.000	0.877

We can see very strong party homophily: Democratic members mostly interact with Democrats (0.874), and Republican members mostly interact with Republicans (0.877). Cross-party interaction is much lower, around 0.12 in both directions. The “Independent” category is too small to interpret. As we can see below, there does not seem to be much difference in these homophily structures between the house and senate.

```
# nominal assortativity
g_clean <- subgraph(g, which(!is.na(V(g)$party)))
party_int <- as.integer(as.factor(V(g_clean)$party))
assortativity_nominal(g_clean, types = party_int, directed =
  ↪ TRUE)
```

```
[1] 0.7150204
```

```
# Homophily by party by chamber
el$from_chamber <- V(g)$chamber[match(el$from, V(g)$name)]

for (ch in c("House", "Senate")) {
  el_ch <- el[!is.na(el$from_chamber) & el$from_chamber == ch, ]
  mix <- xtabs(weight ~ from_party + to_party, data = el_ch)
  mix_prop <- mix / rowSums(mix)
  cat(ch, "\n", knitr::kable(round(mix_prop, 3), format =
    ↪ "pipe"), sep = "\n", "\n")
}
```

House

	Democrat	Independent	Republican
Democrat	0.879	0.001	0.120
Independent	0.985	0.000	0.015
Republican	0.116	0.000	0.884

Senate

	Democrat	Independent	Republican
Democrat	0.826	0.003	0.170

Republican	0.201	0.000	0.799
------------	-------	-------	-------

Left-right ideology assortativity is similarly strong as that of party membership (0.739), meaning members tend to interact with ideologically similar members. Assortativity on the social/racial dimension is weaker (0.225), and age assortativity is essentially negligible (0.101) which means that congress members rarely interact more often with members of their age.

```
g_clean <- induced_subgraph(g, which(!is.na(V(g)$nominate_dim1)))
  ↪ # ensure only non-na values are assessed

# assortativity
leftrightass <- assortativity(g_clean, V(g_clean)$nominate_dim1,
  ↪ directed = TRUE)
socialracialass <- assortativity(g_clean,
  ↪ V(g_clean)$nominate_dim2, directed = TRUE)
ageass <- assortativity(g_clean, V(g_clean)$age24, directed =
  ↪ TRUE)

assort_table <- data.frame(dimension = c("Left-right",
  ↪ "Social/racial issues", "Age"), assortativity =
  ↪ c(leftrightass, socialracialass, ageass))

cat(
  knitr::kable(
    assort_table,
    digits = 3,
    col.names = c("Dimension", "Assortativity"),
    format = "pipe"
  ),
  sep = "\n"
)
```

Dimension	Assortativity
Left-right	0.739
Social/racial issues	0.225
Age	0.101

```
# Homophily by age
el$same_party <- !is.na(el$from_party) &
  !is.na(el$to_party) &
  el$from_party == el$to_party

# filter NAs
el_known <- el %>% filter(!is.na(from_party), !is.na(to_party))
# exclude Independents
```

```

el_known <- el_known[el_known$from_party != "Independent" &
  ↪ el_known$to_party != "Independent", ]

# per-node homophily score
homophily_score <- el_known %>%
  group_by(from) %>%
  summarise(total_w = sum(weight), ingroup_w =
    ↪ sum(weight[same_party]), h_score = ingroup_w / total_w)

homophily_score$age <- V(g)$age24[match(homophily_score$from,
  ↪ V(g)$name)]
homophily_score$party <- V(g)$party[match(homophily_score$from,
  ↪ V(g)$name)]

# correlation
cat("Correlation between age and
  ↪ homophily:\n", cor(homophily_score$h_score,
  ↪ homophily_score$age, use = "complete.obs"), "\n\n")

```

Correlation between age and homophily:
-0.03806106

```

# by party --> democrats weakly positive (+0.04), republicans
  ↪ kind of substantially negative (-0.128)
homophily_score %>%
  group_by(party) %>%
  summarise(`Correlation Age <-> Party Homophily` = cor(h_score,
    ↪ age, use = "complete.obs")) %>%
  mutate(`Correlation Age <-> Party Homophily` =
    ↪ round(`Correlation Age <-> Party Homophily`, 3)) %>%
  { cat(knitr::kable(., format = "pipe"), sep = "\n") }

```

party	Correlation Age <-> Party Homophily
:-----	-----:
Democrat	0.058
Republican	-0.138

Here, we are looking at whether the party homophily of individuals is higher or lower depending on their age. The correlation is very small and slightly negative (-0.038) overall, suggesting that older and younger members do not differ much in their tendency to interact within party. By party, the relationship is weakly positive for Democrats (0.058) and modestly negative for Republicans (-0.138), indicating that older Republican members are somewhat less party-homophilous in their tweet interactions.

Visualization

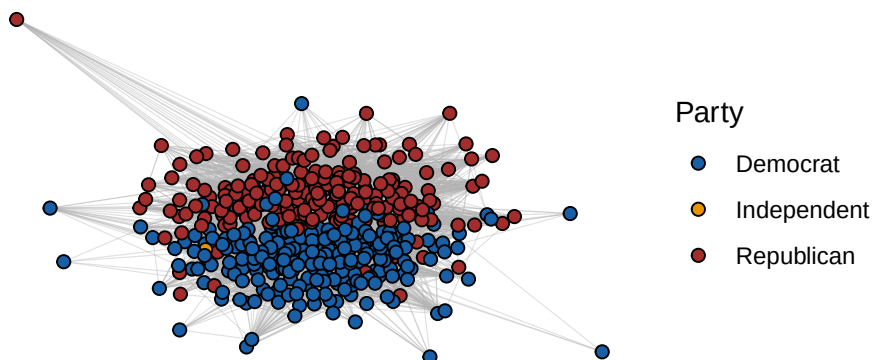
```
g_party <- subgraph(g, which(!is.na(V(g)$party)))
no_of_nodes <- vcount(g_party)

set.seed(7341)

# stress layout with inverted edge weights
ggraph(g_party, layout = "stress", weights = 1 /
  ↪ E(g_party)$weight) +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    ↪ edge_width = 0.2) +
  geom_node_point(aes(fill = party), shape = 21, size = 2) +
  scale_fill_manual(
    values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
      "Independent" = "#E69500"),
    name = "Party"
  ) +
  theme_graph(base_family = "") +
  labs(title = "Stress Layout Overview",
    subtitle = paste0("Nodes with known party only (n = ",
      ↪ no_of_nodes, ")"))
```

Stress Layout Overview

Nodes with known party only (n = 428)



```
set.seed(7341)

# backbone layout needs an undirected graph so we collapse edges
↪ using the mean of in and out
g_party_un <- as_undirected(g_party, mode = "collapse",
  ↪ edge.attr.comb = "mean")
```

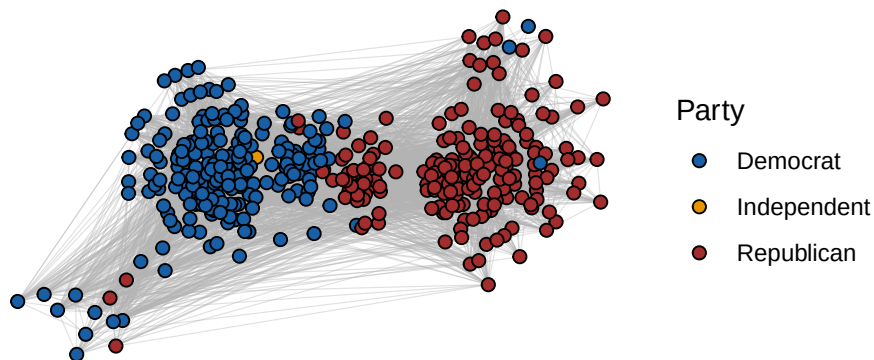
```

ggraph(g_party_un, layout = "backbone") +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    ↪ edge_width = 0.2) +
  geom_node_point(aes(fill = party), shape = 21, size = 2) +
  scale_fill_manual(
    values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
      "Independent" = "#E69500"),
    name = "Party"
  ) +
  theme_graph(base_family = "") +
  labs(title = "Backbone Layout Overview",
    subtitle = paste0("Nodes with known party only (n = ",
      ↪ no_of_nodes, ")"))

```

Backbone Layout Overview

Nodes with known party only (n = 428)



```

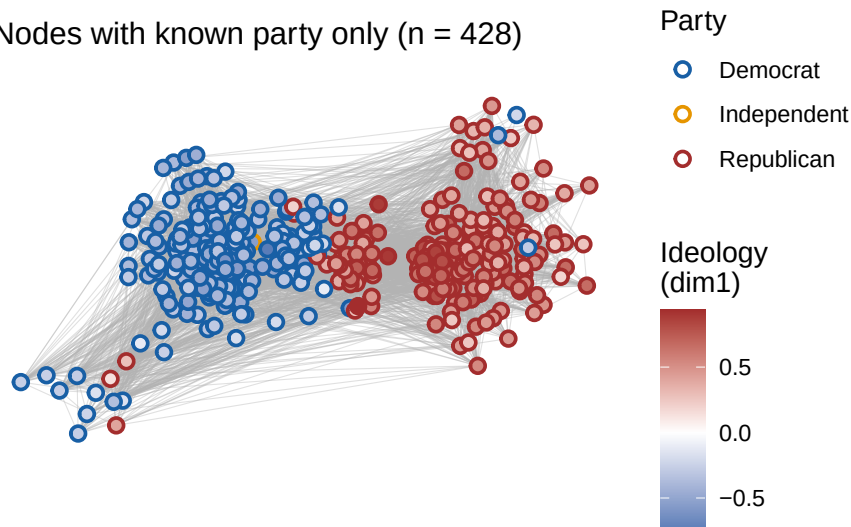
# fill is a gradient by ideology dim1, party is the color contour
ggraph(g_party_un, layout = "backbone") +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    ↪ edge_width = 0.2) +
  geom_node_point(aes(fill = nominate_dim1, colour = party),
    ↪ shape = 21, size = 2, stroke = 1) +
  scale_fill_gradient2(
    low = "#185FA5",
    mid = "white",
    high = "#A32D2D",
    midpoint = 0,
    name = "Ideology\n(dim1)"
  ) +

```

```
scale_colour_manual(
  values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
    "Independent" = "#E69500"),
  name = "Party"
) +
theme_graph(base_family = "") +
labs(title = "Backbone Layout Overview with Ideology",
  subtitle = paste0("Nodes with known party only (n = ",
    no_of_nodes, ")"))
```

Backbone Layout Overview with Ideology

Nodes with known party only (n = 428)



Both layouts confirm strong party homophily with few cross-party exceptions. The single independent node interacts largely with Democrats as expected. Backbone also shows a divide both in the Democratic and Republican party. The latter divide acts as a bridge between the parties. The inclusion of ideology (dim1) also shows according clustering, with higher values closer together, particularly for the Republicans.

```
max_node <- which.max(degree(g, mode = "out"))
cat("Username :", V(g)$username[max_node], "\n")
```

Username : SpeakerPelosi

```
cat("Full name:", V(g)$full_name[max_node], "\n")
```

Full name: Nancy Pelosi

```
cat("Party :", V(g)$party[max_node], "\n")
```

Party : Democrat

```
cat("Age      :", V(g)$age24[max_node], "\n")
```

Age : 84

```
cat("Out-degree:", degree(g, mode = "out")[max_node], "\n")
```

Out-degree: 210

```
cat("Out-strength:", round(strength(g, mode = "out")[max_node],  
  ↪ 4), "\n")
```

Out-strength: 0.9436

```
focal_name <- V(g)$name[V(g)$username == "SpeakerPelosi"]  
ego_g <- make_ego_graph(g, order = 1,  
  nodes = which(V(g)$name ==  
    ↪ focal_name),  
  mode = "all")[[1]]  
focal_idx_ego <- which(V(ego_g)$name == focal_name)  
  
# Top 10 out-edges  
eligible_eids <- as.integer(incident(ego_g, focal_idx_ego, mode =  
  ↪ "out"))  
top10_out_idx <-  
  ↪ eligible_eids[order(E(ego_g)$weight[eligible_eids],  
  ↪ decreasing = TRUE)][1:10]  
ego_out <- subgraph_from_edges(ego_g, eids = top10_out_idx,  
  ↪ delete.vertices = TRUE)  
ego_out <- induced_subgraph(ego_out,  
  ↪ which(!is.na(V(ego_out)$party)))  
focal_idx2 <- which(V(ego_out)$name == focal_name)  
  
ggraph(ego_out, layout = "focus", focus = focal_idx2) +  
  draw_circle(col = "grey80", use = "focus", max.circle = 1) +  
  geom_edge_link0(aes(edge_width = weight), edge_colour =  
    ↪ "grey25", edge_alpha = 0.5) +  
  scale_edge_width_continuous(name = "Transmission\nprobability",  
    ↪ range = c(0.3, 2.5)) +  
  geom_node_point(aes(fill = party), shape = 21, size = 3) +  
  scale_fill_manual(  
    values = c("Democrat" = "#185FA5", "Republican" =  
      ↪ "#A32D2D",  
      "Independent" = "#534AB7"),  
    na.value = "grey80",  
    name = "Party"
```

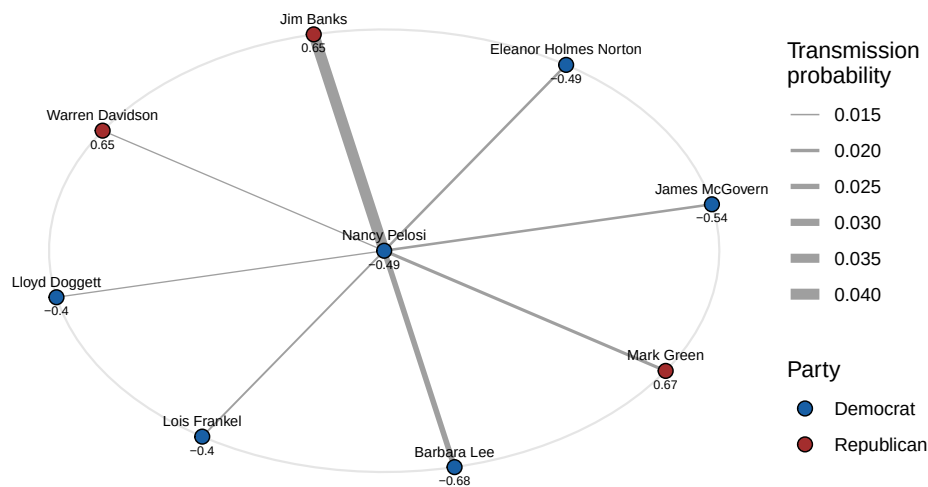
```

) +
geom_node_text(aes(label = full_name), size = 2.5, vjust = -1)
↪ +
geom_node_text(
  aes(label = round(nominate_dim1, 2)),
  size = 2.2, vjust = 2, colour = "black"
) +
theme_graph(base_family = "") +
coord_cartesian(clip = "off") +
labs(title = paste("Ego network:",
  ↪ V(ego_out)$full_name[focal_idx2]),
  subtitle = "Top 10 out-edges by transmission probability.
  ↪ \nValues below nodes indicate DW-Nominate dimension
  ↪ 1.")

```

Ego network: Nancy Pelosi

Top 10 out-edges by transmission probability.
Values below nodes indicate DW-Nominate dimension 1.



We look at the top 10 edges of an example node, Nancy Pelosi, in an ego network. She has the highest out-degree of all nodes. Despite the majority of the nodes following party and ideology homophily, there are major exceptions, particularly with Jim Banks who has an ideological distance of 1.14 to Pelosi. This is likely a result of conflict interaction rather than collaboration. This illustrates that aggregate homophily patterns can coexist with meaningful individual-level violations, particularly among high-visibility leadership figures.

```

el <- igraph::as_data_frame(g, what = "edges")
el$from_party <- V(g)$party[match(el$from, V(g)$name)]

```

```

el$to_party      <- V(g)$party[match(el$to, V(g)$name)]
el$from_dim1     <- V(g)$nominate_dim1[match(el$from, V(g)$name)]
el$to_dim1       <- V(g)$nominate_dim1[match(el$to, V(g)$name)]
el$ideo_distance <- abs(el$from_dim1 - el$to_dim1)

assort_by_bin <- el %>%
  filter(!is.na(from_party), !is.na(to_party),
    ↪ !is.na(ideo_distance),
    from_party != "Independent", to_party != "Independent")
  ↪ %>%
  mutate(
    edge_type = ifelse(from_party == to_party, "Within-party",
    ↪ "Cross-party"),
    distance_bin = cut(ideo_distance, breaks = seq(0, 2, by =
    ↪ 0.25),
    include.lowest = TRUE)
  ) %>%
  group_by(edge_type, distance_bin) %>%
  summarise(n_edges = n(), mean_weight = mean(weight), .groups =
    ↪ "drop") %>%
  filter(!is.na(distance_bin)) %>%
  tidyr::complete(edge_type, distance_bin)

```

Looking at mean edge weights by ideological distance, we high interaction within parties on low ideological distance. Similar high interaction persists across parties even as distance increases. Only at the highest distance bin does it decrease, meaningful interaction across parties exists up to a threshold. We might expect interaction to increase at extreme ideological differences due to conflict, but we do not observe this. It suggests that conflict-driven engagement, while present, does not dominate aggregate interaction patterns.

Conclusion

Results consistently confirm strong symmetric partisan homophily, across both parties and chambers. Ideological assortativity on the left-right dimension reinforces this pattern, while the social-racial dimension and age produce much weaker effects. Network visualizations confirm partisan and ideological clustering. Exceptions exist, though these ties likely reflect conflict interaction rather than cross-party collaboration.

Key limitations include 47 unmatched nodes, the limitation to the 117th Congress, and the inability to distinguish collaborative from adversarial interactions in the transmission probability metric. Future research could incorporate interaction type classification to separate confrontational from collaborative ties, integrate committee co-membership data as a structural

covariate, or extend the analysis across multiple congresses to assess whether homophily has intensified alongside broader political polarization.

References

- McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. “Birds of a Feather: Homophily in Social Networks.” *Annual Review of Sociology* 27 (1): 415–44. <https://doi.org/10.1146/annurev.soc.27.1.415>.
- Newman, M. E. J. 2003. “Mixing Patterns in Networks.” *Physical Review E* 67 (2): 026126. <https://doi.org/10.1103/PhysRevE.67.026126>.